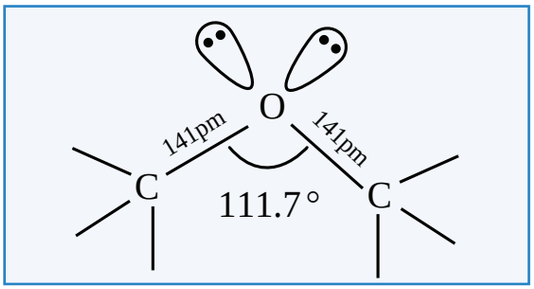
Structure of functional group



The structure of ethereal oxygen which is attached to two alkyl groups is similar to the structure of -O-H group of alcohol. The oxygen atom is sp^3 hybridized. Two sp^3 hybridized orbitals of oxygen linearly overlap with two sp^3 hybrid orbitals of the carbon which are directly attached to the oxygen forming two C-O 'σ ' bonds. The C-O-C bond angle is slightly greater than the tetrahedral bond angle due to the repulsive interaction between the two bulkier alkyl groups.

IUPAC System:

Let us recall the naming of ethers according to IUPAC nomenclature.

Compound (Common Name, Structural formula, IUPAC Name)	IUPAC Name			
	Prefix with position number	Root used	Primary suffix	Secondary Suffix
Dimethyl ether $CH_3—O—CH_3$ Methoxymethane	Methoxy	Meth	ane	–
Isopropyl methyl ether $CH_3—O—\underset{\displaystyle CH_3}{\underset{\vert}{CH}}—CH_3$ 2-methoxypropane	2-methoxy	prop	ane	–
t-butylmethyl ether $CH_3-O-\underset{\displaystyle CH_3}{\underset{\vert}{\overset{\displaystyle CH_3}{\overset{\vert}{C}}}}-CH_3$ 2-methoxy-2-methyl propane	2-methoxy 2-methyl	prop	ane	–
Methylphenylether(Anisole) $C_6H_5—O—CH_3$ Methoxybenzene	Methoxy	benzene	–	–
Ethylphenylether(phenetole) $C_6H_5—O—CH_2CH_3$ Ethoxybenzene	Ethoxy	benzene	–	–
Diphenylether or phenylether $C_6H_5—O—C_6H_5$ Phenoxybenzene	Phenoxy	benzene	–	–

XII U11-Hydroxy compounds.indd 133 11-01-2021 15:45:01